Begonnen am	Monday, 2. May 2022, 15:31
Status	Beendet
Beendet am	Monday, 2. May 2022, 16:03
Verbrauchte Zeit	32 Minuten 17 Sekunden
Bewertung	**50,00** von 50,00 (**100**%)

Frage **1**

Vollständig

Nicht bewertet

Read this carefully!

General information:

- Not counting this first question, there are 10 problems with 50 points in total.
- Read all problems and possible answers carefully and take your time to think about them.
- Multiple-choice questions: Choosing wrong answers leads to deduction of points within that question. However, you cannot get less than 0 points on a question.
- Numerical questions: Wrong answers do not lead to point deductions.
- For numerical answers, use "." for the decimal point. In case you get an error message or the question is marked as not answered, try using "," instead.
- When it says that you can use a pocket calculator, you may also use the calculator of your operating system.
- You can ask questions during the exam only via the chat window and only directly to the exam supervisor.
- You have 50 minutes for the exam. After this time is expired, your exam will automatically be submitted. There is no overtime.
- Once you have submitted your exam, please state so in the chat window, don't wait for a reply, and then just leave the Zoom meeting.

Statutory declaration:

With clicking on "I agree" below, you agree to this statutory declaration ("Eidesstattliche Erklärung"): "I hereby confirm that I complete this closed-book exam on my own and only using the allowed materials, in particular without any written or electronic notes and without any help from or communication with other people. I understand that violation of this declaration may cause legal consequences."

Wählen Sie eine Antwort:

- (x) a. I agree
- ( ) b. I do not agree

Frage **2**

Richtig

Erreichte Punkte 5,00 von 5,00

Which statements about Unsupervised Machine Learning are true?

Multiple answers are possible!

Wählen Sie eine oder mehrere Antworten:

- [ ] a. Underfitting means that a model is adapted to special training characteristics such as noisy measurements or outliers.
- [ ] b. Projection methods project data into a higher-dimensional space while keeping their neighborhoods.
- [x] c. The model selection procedure is called "training" or "learning".
- [x] d. Examples of unsupervised methods are: PCA, ICA, Factor Analysis, k-means clustering, and mixture models.
- [x] e. In parametric models, every parameter (vector) realization represents a model.
- [x] f. k-nearest neighbors is a nonparametric model.
- [ ] g. Generative models do not aim at modeling the data creation process.
- [x] h. Unsupervised Machine Learning alorithms learn patterns from unlabeled data.

Frage **3**

Richtig

Erreichte Punkte 5,00 von 5,00

Which statements about Parameter Estimation and Maximum Likelihood are true?

Multiple answers are possible!

Wählen Sie eine oder mehrere Antworten:

- [x] a. If the data are distributed i.i.d., the likelihood can be written as a product of individual probabilities over the data samples.
- [ ] b. The likelihood and the log-likelihood are maximized for the same parameter value, and at that parameter value $\hat{\mathbf{w}}$, their values coincide: $\mathcal{L}(\hat{\mathbf{w}}) = \ln \mathcal{L}(\hat{\mathbf{w}})$.
- [ ] c. Parameter estimation and the maximum likelihood approach rest on the assumption that the model class is not know and estimated from the data.
- [ ] d. The maximum likelihood estimator is asymptotically unbiased and efficient, but not asymptotically consistent.
- [x] e. The mean squared error can be written as the sum of the variance of the estimator and the squared bias of the estimator.
- [x] f. An estimator $\hat{\mathbf{w}}$ is unbiased if its expected value (over the training set distribution) equals the true parameter: $\mathrm{E}(\hat{\mathbf{w}}) = \mathbf{w}$.
- [x] g. The likelihood $\mathcal{L}(\{\mathbf{x}\}; \mathbf{w})$ is the probability that the model with parameter $\mathbf{w}$ produces the data $\{\mathbf{x}\}$.
- [ ] h. The mean squared error is defined as follows: $\mathrm{mse}(\hat{\mathbf{w}}) = (\hat{\mathbf{w}} - \mathbf{w})^T (\hat{\mathbf{w}} - \mathbf{w})$.

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Frage **4**

Richtig

Erreichte Punkte 5,00 von 5,00

Which statements about Expectation Maximization (EM) and Maximum Entropy are correct?

Multiple answers are possible!

Wählen Sie eine oder mehrere Antworten:

- [x] a. The EM algorithm can be applied to hidden Markov models and mixture models.
- [x] b. In the M-step, the model parameters are updated.
- [ ] c. The E-step increases the upper bound on the log-likelihood.
- [x] d. In EM, one deals with problems that have hidden variables as well as model parameters.
- [ ] e. The entropy of a probability distribution is always larger equal 0 and smaller equal 1.
- [ ] f. Hidden state models in general have analytical solutions for the likelihood.
- [x] g. In the E-step, the hidden variables parameters are updated.
- [x] h. The Maximum Entropy approach requires minimal prior assumptions to assign a-priori-probabilities.

Frage **5**

Richtig

Erreichte Punkte 5,00 von 5,00

Assume you are given two binary random variables X and Y taking only the values 0 and 1 where their joint probability distribution is specified by the table below:

	$Y=0$	$Y=1$
$X=0$	$1/3$	$1/3$
$X=1$	0	$1/3$

This means, that e.g. $P(X=0, Y=0) = 1/3$, etc.

Compute the quantities (marginal probabilities, entropies, and mutual information) below. Use the natural logarithm in your calculations.

You can use a pocket calculator. Round to two decimal places. Number format example: 0.67

- $P(Y=0) =$ 0,33
- $H(X) =$ 0,64
- $H(Y) =$ 0,64
- $I(X;Y) =$ 0,17

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Frage **6**

Richtig

Erreichte Punkte 5,00 von 5,00

Which statements about Principal Component Analysis (PCA) are correct?

Multiple answers are possible!

Wählen Sie eine oder mehrere Antworten:

- [x] a. The first principal component corresponds to the largest eigenvalue in the eigendecomposition of the sample covariance matrix.
- [ ] b. The singular values from the singular value decomposition of the data matrix $\mathbf{X}$ are equal to the eigenvalues of the eigendecomposition of $\mathbf{X}^T\mathbf{X}$.
- [ ] c. The PCA solution is unique, i.e., there is only one solution.
- [x] d. Eigenvectors can be computed iteratively with Oja's rule.
- [x] e. The sample variance of the k-th projection is equal to the k-th eigenvalue of the sample covariance matrix.
- [x] f. The matrix of eigenvectors $\mathbf{U}$ is an orthogonal matrix.
- [x] g. The sample covariance matrix is symmetric and positive-semidefinite.
- [ ] h. PCA transforms data into a new coordinate system such that the data has the largest mean value along the first coordinate (first principal component), the second largest mean along the second, and so on.

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Frage **7**

Richtig

Erreichte Punkte 5,00 von 5,00

PCA:

You are given 4 data samples, each of dimension 3. The data matrix reads:

$$\mathbf{X} = \begin{pmatrix} 1 & 1 & 1 \\ 2 & 2 & 2 \\ 3 & 3 & 3 \\ 4 & 4 & 4 \end{pmatrix}$$

The eigenvectors of $\mathbf{X}^T\mathbf{X}$ are:

$$\mathbf{u}_1 = \frac{1}{\sqrt{3}}(1,1,1)^T$$
$$\mathbf{u}_2 = \frac{1}{\sqrt{2}}(-1,0,1)^T$$
$$\mathbf{u}_3 = \frac{1}{\sqrt{2}}(-1,1,0)^T$$

These eigenvectors are already normalized and they are already correctly ordered corresponding to eigenvalues in descending order. (Just ignore that the second and third eigenvector are not orthogonal. This is due to the fact that the data matrix only has rank 1.)

Compute the PCA projection (matrix) $\mathbf{Y}$.

Enter the components of the projected second feature vector $\mathbf{y}_2$ into the fields below.

You can use a pocket calculator. Round to two decimal places. Number format example: 0.67

- $y_2^{(1)} =$ 3,46
- $y_2^{(2)} =$ 0
- $y_2^{(3)} =$ 0

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Frage **8**

Richtig

Erreichte Punkte 5,00 von 5,00

Tick the correct statements about Kernel PCA.

We stick to the notation from the lecture, i.e. the data points are denoted by $\mathbf{x}_1, \ldots \mathbf{x}_n \in \mathcal{X} = \mathbb{R}^m$, the kernel by $k : \mathcal{X} \times \mathcal{X} \to \mathbb{R}$, and the feature map by $\Phi : \mathcal{X} \to \mathcal{H}$. The feature covariance matrix is defined by $\mathbf{C} = \frac{1}{n} \sum_{i=1}^{n} \Phi(\mathbf{x}_i)\Phi(\mathbf{x}_j)^T$ and the Gram matrix $\mathbf{K}$ by $K_{ij} = \Phi(\mathbf{x}_i)^T \Phi(\mathbf{x}_j)$ for $i, j = 1, \ldots, n$.

Wählen Sie eine oder mehrere Antworten:

- ☐ a. A disadvantage of Kernel PCA is that Φ always needs to be computed explicitely, which is computationally costly.
- ☐ b. In general, any nonnegative and symmetric function $k : \mathcal{X} \times \mathcal{X} \to \mathbb{R}$ defines a kernel.
- ☑ c. To compute the projection of a data point $\mathbf{x}$ onto the eigenvectors of $\mathbf{K}$, you need to center $\mathbf{x}$ and $\mathbf{K}$, as well as to normalize the eigenvectors of $\mathbf{K}$.
- ☑ d. The main task of kernel PCA is to solve the eigenvalue problem of $\mathbf{C}$ in feature space. This can be achieved by solving the eigenvalue problem of $\mathbf{K}$ in a Euclidean space.
- ☑ e. In general, Φ is a nonlinear map that maps into a higher dimensional feature space $\mathcal{H}$.
- ☑ f. Linear PCA can be seen as a special case of kernel PCA by using $k(\mathbf{x}, \mathbf{y}) = \mathbf{x}^T\mathbf{y}$ as the associated kernel function.
- ☐ g. For kernel PCA, it is essential to center $\mathbf{K}$ before computing its eigendecomposition, since the eigenvalues of $\mathbf{K}$ can be in general be negative, which leads to inconclusive results.
- ☐ h. $\mathbf{K}$ and $\mathbf{C}$ always have the same dimensionality and the same rank.

Frage **9**

Richtig

Erreichte Punkte 5,00 von 5,00

Which statements about Independent Component Analysis (ICA) are correct?

Multiple answers are possible!

Wählen Sie eine oder mehrere Antworten:

- ☐ a. Maximizing the negentropy maximizes the mutual information.
- ☑ b. In general, ICA components are neither ranked nor orthogonal.
- ☐ c. Non-Gaussianity can be measured by the second cumulant.
- ☑ d. Criteria for independence are mutual information between components and non-Gaussianity of components.
- ☐ e. The ICA solution is unique.
- ☐ f. For ICA, one needs at least as many sources as observations.
- ☐ g. Infomax maximizes the mutual information between components.
- ☑ h. Fast ICA relies on whitening and rotating the data.

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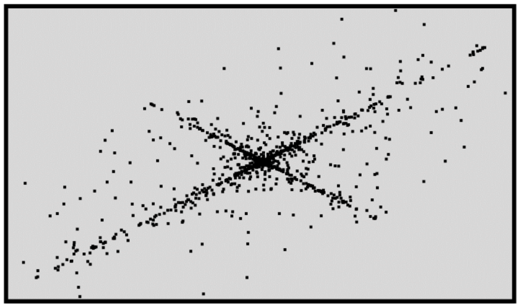
Frage **10**

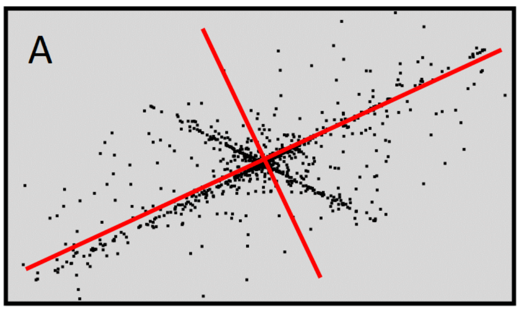
Richtig

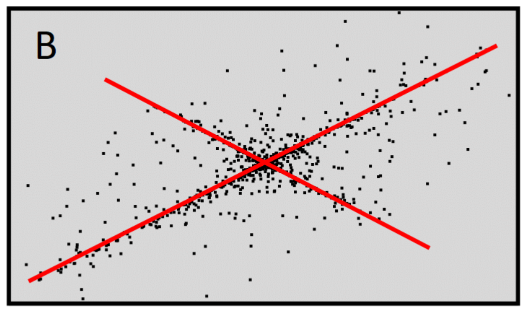
Erreichte Punkte 5,00 von 5,00

Tick the correct statements related to the pictures below:

- ICA and PCA are applied to the dataset shown in the picture in the first row (only black dots), leading to the outcomes A and B. Then picture A is the result of PCA .

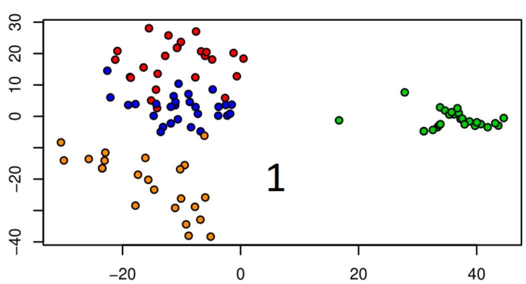


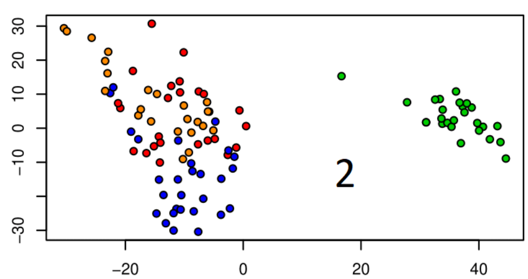




- Pictures 1 and 2 below show the result of an application of PCA to a certain dataset with 4 features in a new coordinate system given by certain principal components (PCs). The 4 different features are represented by the different colors of the dots. The PCs are ranked in the usual way and numbered accordingly. Tick the statement, which is more likely:

Picture 1 shows a plot of the dataset in the coordinate system PC2 vs PC1, whereas picture 2 visualizes the situation ir





- The picture on the left shows the original dataset where the other pictures a and b show the results of an application of the EM-algorithm with a mixture of two Gaussians to the dataset. Tick the statement, which is more likely:

Picture a is the result after doing 5 iterations, whereas picture b is the result after 20 iterations.

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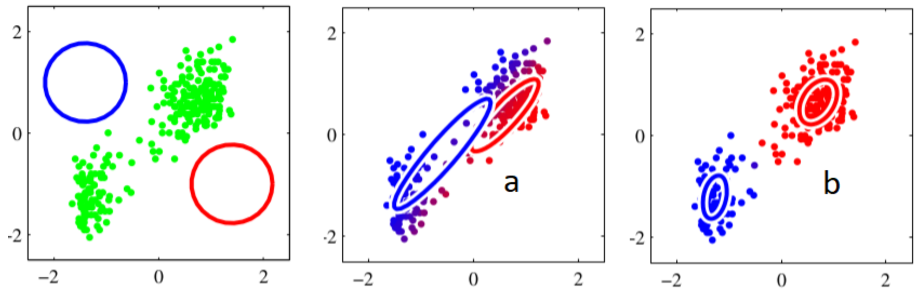

Frage **11**

Richtig

Erreichte Punkte 5,00 von 5,00

Assume you have two independent random variables X and Y, which are uniformly distributed on $[-1, 1]$, i.e.

$$p_X(x) = p_Y(x) = \begin{cases} \frac{1}{2} & \text{if } x \in [-1, 1] \\ 0 & \text{else.} \end{cases}$$

As in the lecture, let $\kappa_n(X)$ denote the n-th order cumulant of a random variable X. Compute the quantities and answer the questions below.

Hint: Recall that $\kappa_4(X) = \mathrm{E}(X^4) - 3(\mathrm{E}(X^2))^2$ if X is centered.

You can use a pocket calculator. Round to three decimal places. Number format example: 0.667

- $\kappa_4(X) =$ -0,133
- $\kappa_4(2X) =$ -2,133
- $\kappa_4(X+Y) =$ -0,267
- X is sub Gaussian

◄ Lecture Notes ML unsupervised

Direkt zu:

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